09-May-2026

Agenda:

Feature Engineering - 7

→ Feature Selection

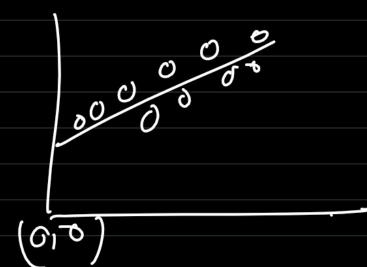
→ Imbalance data

Machine learning:

$y = mx \rightarrow y = mx + c \rightarrow$

value

intercept



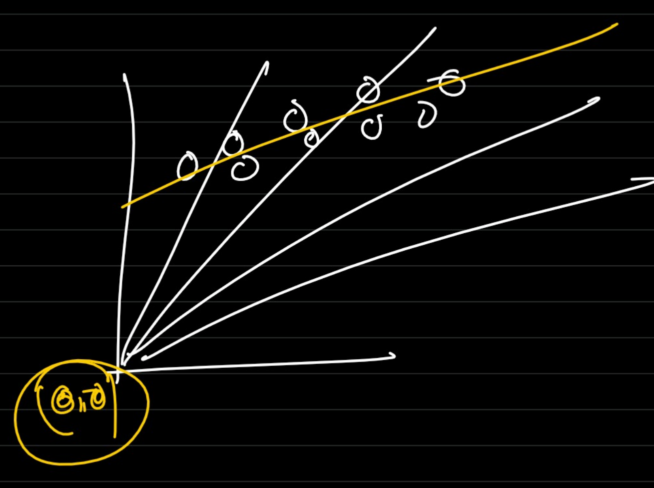
col	x_1	x_2	x_3	x_4
1				
1				
1				
1				
1				
1				
1				

1401 + rows

→ MC

$x_1 \; x_2 \; x_3 \; x_4 \rightarrow m_1 \; m_2 \; m_3 \; m_4$

C-Col



$$y_{pred} \leftarrow m_1x_1 + m_2x_2 + m_3X_3 + m_4X_4 + C$$

$y = 100$

$y_{pred} = 200$

$$(y - y_{pred}) = -100 \longrightarrow \begin{matrix} m_1 \\ m_2 \\ m_3 \\ m_4 \\ c \end{matrix}$$

$$[\,1 \;\; x_1 \;\; x_2 \;\; x_3 \;\; x_4\,] \begin{bmatrix} c \\ m_1 \\ m_2 \\ m_3 \\ m_4 \end{bmatrix} \longrightarrow y_{pred}$$

$\boxed{}$ = fit _ transfm

fit ↓ learn

FS → model(intercept)

→ model[intercept + feature[0]] →

→ model(intercept + feature[0] + feature[1])

feature(1)

select

Selected-features = [A, B]

remaining-features = (~~A~~, ~~B~~, C, D)

const + A → p(A) → 0.01
const + B → p(B) → 0.02
const + C → p(C) → .03
const + D → p(D) → .07

min_pvalue

while remaining-features :

p_vals = { }

for f in remaining-feature:

ML → (y, [const + f + selected]) → const + A + B →
const + A + C → pvalue
const + A + D →

p_vals [f] = p-value

const + A + B → C
→ D

p-val_min < threshold :

→ (f) → selected

const + A + B + C → D
→ 0.08

→ remaining (f)

break

R^2 (coefficient of determination)

→ uses the mean of the features & compares it with predict values.

→ measures how much better a model is than simply guessing the average value

Actual | Pred | Actual-mean

x_1 x_2 x_3 x_4 Height (y) ($m_1x_1 + m_2x_2 + m_3x_3 + m_4x_4 + c$)

→ High
→ ranged
→ low

error = High

errors =

x_1, x_2	Height (y)	$\hat{y}$	|error|	(mean) naive-model	|error|
-,-	170	210	40	190	20
-,-	180	310	130	190	10
-,-	190	400	210	190	0
-,-	200	120	80	190	10
-,-	210	110	100	190	20
			560		60

$x_1, x_2 \rightarrow y \rightarrow$

$x_1, x_2 \rightarrow y$

$x_1, x_2 \rightarrow \boxed{\dfrac{\text{error}(y - y_{mean})}{\text{error}(y - y_{pred})}}$ model is better than mean or not

$R^2 \rightarrow$ explains the %tage of the variance in the dependent variable (target) explained by independent variables (features)

$$R^2 = 1 - \frac{SS_{res} \rightarrow \text{sum of residuals (Actual - predicted)}}{SS_{total} \rightarrow \text{sum of square difference (Actual - mean of Actual)}}$$

R^2 of 1 means model perfectly explains all the variance

R^2 of 0 means model explains none of the variance

$1 - \dfrac{2.5}{5.0} = 0.5$ $\qquad 1 - \dfrac{5.0}{2.5} = -1$

bad $\leftarrow$ 0 $\rightarrow$ 1 $\rightarrow$ good

-1 $\leftarrow$ $\qquad \rightarrow$

$\boxed{0-1}$

$X \quad y \quad y_{pred} \quad y_{mean}$

R^2

x_1

A
B
C

A

$\boxed{const + X}$

$const + x_1 + x_2$

$\boxed{const + x_1 + x_2 + x}$

$\boxed{const + A} + B$

$+ C$

RFE:

The bigger the coefficient of linear equation are the (m_1, m_2, m_i) more influence their respective feature have. (x_1, x_2, x_3, x_n)

- RFE works like BE but without p-value & R^2
- This uses model coefficients
- linear model, trees

linear model ↓ coefficient

trees ↓ impurity

RFE (top_feature_to_select = 5)

5 → coefficient → higher value

$$y = m_1 x_1 + m_2 x_2 + c$$

$$y - y_{pred} = 0.1\, x_1 + 0.89\, x_2 + 0.1$$

err → minimize

↳ |e| → err

↳ |e|

↳ |e|

↳ |e|

Best → m1 m2

→ −0.6 2

↓

m2

↓

RFE [top = 2] → m1 m2 → 1

top = 1] → m2 → 1

→ m1 → 2